

Phase 1 – Brainstorming & Ideation

AI WeatherWise Project

Project: AI WeatherWise

1. Project Idea

AI WeatherWise is a RESTful backend platform using Node.js and Express.js, MongoDB/Mongoose, and Google Gemini AI to provide intelligent weather summaries and personalized recommendations.

2. Problem Statement

- Difficulty tracking and managing multiple favorite weather locations efficiently.
- Lack of personalized recommendations for outdoor activities or clothing.
- Need for natural-language weather summaries instead of only raw weather metrics.
- Need for a secure system to store and sync preferences across devices.

3. Proposed Solution

The system centralizes weather-data retrieval and AI orchestration, manages users and favorite locations, and provides customized insights and recommendation workflows through Gemini.

4. Objectives

- Provide real-time weather information through APIs.
- Manage favorite locations.
- Generate AI-driven weather summaries and recommendations.
- Provide secure authentication and role-based access.
- Maintain data using MongoDB and Mongoose.

5. Users / Roles

- Admin – user management, API health/system logs and database-integrity oversight.
- Registered User – favorite locations, weather requests, AI insights and profile management.
- Public Weather Application User – consumes public weather endpoints without an account.